सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

NCVET
कौशल गुणवत्ता प्रगति

N·S·D·C
RE-IMAGINE FUTURE

Qualification Pack

Technician: Distribution Transformer Operation and Maintenance

QP Code: PSS/Q2406

Version: 2.0

NSQF Level: 4

Power Sector Skill Council || Plot No. 4, Institutional Area, CBIP Building, Malcha Marg, Chanakyapuri New Delhi-110021 || email:akbhagatin@yahoo.co.in

Contents

Qualification Pack

PSS/Q2406: Technician: Distribution Transformer Operation and Maintenance

Brief Job Description

The incumbent in the job is responsible for checking, testing, operation, repair, overhaul and maintenance of distribution transformer (DT) of rating 11/0.433 kV.

Personal Attributes

The job requires the individual to have physical strength and ability to read, write and communicate, stand for long working hours. The individual should be mentally strong.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. PSS/N1338: Apply basic health And safety practices for power related work
2. PSS/N1336: Working effectively with others
3. DGT/VSQ/N0101: Employability Skills (30 Hours)
4. PSS/N2406: Testing and inspection of various faults in distribution transformer
5. PSS/N2417: Operation and Maintenance of Distribution Transformer

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution
Country	India
NSQF Level	4
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3113.0200

Minimum Educational Qualification & Experience	Completed 2nd year of the 3-year diploma after 10 OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant experience OR 10th grade pass plus 2-year NTC (/NAC in relevant trade) OR 12th grade Pass ((Science))
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	2.0
Reference code on NQR	QG-04-PW-04250-2025-V2-PSSC
NQR Version	2

PSS/N1338: Apply basic health And safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)

PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)

PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment

PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)

PC5. ensure positive isolation of electrical equipment and system as per the given standards

PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.

PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)

PC8. use proper scaffolds, elevated platforms and ladders for working at height

PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages

PC10. store flammable materials and machine lubricating oil safely and correctly

PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO2, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read & write an accident/ incident report in local language and/or English

GS2. read and comprehend basic content to read labels, charts, signages, manuals etc.

GS3. question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions

GS4. remain congenial while discussing and debating issues with co-workers

GS5. observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice

GS6. plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

GS7. build customer relationships and use customer centric approach

GS8. seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Follow workplace health and safety practices	**25**	**36**	**-**	**-**
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
Follow fire safety practices	**6**	**12**	**-**	**-**
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
Follow emergencies, rescue and first-aid procedures	**8**	**13**	**-**	**-**
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	**39**	**61**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1338
NOS Name	Apply basic health And safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required

PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt

PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency

PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism

PC5. listen attentively and respond appropriately while interacting with others at work

PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)

PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

PC8. respect gender diversity in the workplace

PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive

PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture

PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
KU2. reporting structure, inter-dependent functions, lines and procedures in the work area
KU3. relevant people and their responsibilities within the work area
KU4. escalation matrix and procedures for reporting work and employment related issues
KU5. key elements of active listening and assertive communication
KU6. importance of teamwork in organizational and individual success
KU7. ethics and disciplined behavior for professional success
KU8. expressing and addressing grievances appropriately and managing interpersonal conflict effectively
KU9. harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
KU10. organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
GS2. read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
GS3. interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
GS4. display active listening skills while interacting with co-workers and other in the workplace
GS5. follow organization’s rule-based decision-making process
GS6. use appropriate planning to maintain a smooth relationship with fellow team members
GS7. meet customer requirements
GS8. critically evaluate operation parameters in relation to system normality

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Work effectively with others	**18**	**32**	**-**	**-**
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
Respect diversity	**16**	**24**	**-**	**-**
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	**34**	**56**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

KU10. how to compute income and expenses
KU11. importance of maintaining safety and security in financial transactions
KU12. different legal rights and laws
KU13. how to operate digital devices and applications safely and securely
KU14. ways to identify business opportunities
KU15. types of customers and their needs
KU16. how to apply for a job and prepare for an interview
KU17. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively using appropriate language
GS2. behave politely and appropriately with all
GS3. perform basic calculations
GS4. solve problems effectively
GS5. be careful and attentive at work
GS6. use time effectively
GS7. maintain hygiene and sanitisation to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**1**	**3**	**-**	**-**
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
Communication Skills	**1**	**1**	**-**	**-**
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**1**	**-**	**-**
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
Financial and Legal Literacy	**3**	**4**	**-**	**-**
PC9. use various financial products and services safely and securely	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
Essential Digital Skills	**4**	**6**	**-**	**-**
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
Entrepreneurship	**3**	**5**	**-**	**-**
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
Customer Service	**2**	**2**	**-**	**-**
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**1**	**3**	**-**	**-**
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

PSS/N2406: Testing and inspection of various faults in distribution transformer

Description

This NOS unit covers the inspection of a Distribution Transformer in a safe and effective manner.

Scope

The scope covers the following :

- Identifying faults and their root cause in a distribution transformer (DT)

Elements and Performance Criteria

Identifying faults and their root cause in distribution transformer (DT)

To be competent, the user/individual on the job must be able to:

PC1. prepare check list of parameters for consideration during testing and inspection of faulty distribution transformer

PC2. identify faults by visually inspect the general appearance, leakage of oil, and physical condition such as rust on body and radiators of DT

PC3. verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections

PC4. carry out insulation-resistance test to identify grounding and shorting faults in the connections

PC5. check silica gel breather for the oil level in oil cap, breathing holes in silica gel breather and color of silica gel

PC6. check condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket, gasket joints and flanges for any damage, flash and hair crack

PC7. identify faults arising due to primary winding burnt (one phase, two phase or complete), braze /solder of LT winding joints melted, over heat, open circuit in internal wiring etc.

PC8. inspect excess humming noise in DT and identify loose fitting of silicon mixed steel alloys laminated core joints

PC9. maintain records of DT basic information like serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the DT performance history

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. process documentation, inspection and testing standards and procedures followed in the organization

KU2. related power system aspects like ratings and various types of DT

KU3. working of a DT, its component, accessories and their functioning

KU4. difference between dry type and oil immersed transformer and their usage at site

KU5. various material / parts / accessories required for maintenance like insulating oil, type of core, winding material, bushings, cable and conductor, cabling box, cooling radiators, conservators, oil gauges, valves, explosion vents or pressure release devices, sealing gaskets, temperature indicators, poles, insulators and fuses etc.

KU6. use of tools and kits required for testing, repair and maintenance : OLTC Continuity & Resistance Measurement. (Test: Dissolve Gas Analyzer kit, temperature monitor device, Partial Discharge Test Set, Flash Point Test Set of Oil, discharge rod, chain pulley, tripod, crane, hoist, force pulley with sling, tommy bar, crimping machine, drilling machine, meggar, tong tester etc.)

KU7. procedure and technical requirements for testing, repair and maintenance of the distribution transformer

KU8. causes and methods of repairing major faults in a transformer

KU9. how to test the performance and condition of distribution transformer

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. write reports and observations related to work in English/regional language

GS2. attentively listen and comprehend the information given by the supervisor/team members

GS3. read and interpret the equipment operational documents

GS4. discuss task lists and job requirements with supervisor and co-workers

GS5. follow organization rule-based decision making process in consultation of the supervisor

GS6. analyze the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist

GS7. recognize a workplace problem and take suitable action to resolve it

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Identifying faults and their root cause in distribution transformer (DT)	**22**	**78**	**-**	**-**
PC1. prepare check list of parameters for consideration during testing and inspection of faulty distribution transformer	3	5	-	-
PC2. identify faults by visually inspect the general appearance, leakage of oil, and physical condition such as rust on body and radiators of DT	2	7	-	-
PC3. verify connections of HT/LT side and inspect bolt/lugs and solder of electrical connections	3	8	-	-
PC4. carry out insulation-resistance test to identify grounding and shorting faults in the connections	2	7	-	-
PC5. check silica gel breather for the oil level in oil cap, breathing holes in silica gel breather and color of silica gel	2	11	-	-
PC6. check condition of OLTC, leakage of oil in bushing collar, porcelain insulator bushing, gasket, gasket joints and flanges for any damage, flash and hair crack	3	13	-	-
PC7. identify faults arising due to primary winding burnt (one phase, two phase or complete), braze /solder of LT winding joints melted, over heat, open circuit in internal wiring etc.	3	13	-	-
PC8. inspect excess humming noise in DT and identify loose fitting of silicon mixed steel alloys laminated core joints	2	8	-	-
PC9. maintain records of DT basic information like serial number, diagram, rating plate etc. and defects/repair showing diagnostic records for assessing the DT performance history	2	6	-	-
NOS Total	**22**	**78**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2406
NOS Name	Testing and inspection of various faults in distribution transformer
Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution
NSQF Level	4
Credits	5.5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

PSS/N2417: Operation and Maintenance of Distribution Transformer

Description

This unit covers the Operation and Maintenance of Distribution Transformer of all types of Distribution transformer including repair , Protection ,safety,emergency handling and Cyber security.

Scope

The scope covers the following :

- The responsibilities include repairing, testing, and checking overhauled distribution transformers before delivery, along with ensuring their smooth operation and maintenance. The role also covers implementing protection and safety measures for transformers, as well as responding promptly to transformer-related faults, emergencies, and issues linked to digital equipment or cybersecurity concerns.

Elements and Performance Criteria

Repairing of defective distribution transformer

To be competent, the user/individual on the job must be able to:

PC1. check that all required tools and kits are calibrated and in good condition

PC2. prepare work area for the standard repairing procedure

PC3. identify and collect adequate spare parts to replace the faulty parts

PC4. take oil samples from tank bottom, tank top, and radiator to conduct the Break-Down Voltage (BDV) test

PC5. remove core and windings from the tank for visual inspection and ensure that they are properly covered, dry and placed safely after removal from tank

PC6. inspect core, primary winding, secondary winding, primary terminal connections, secondary terminal connections, insulation (fish paper, empire tape/cloth, wooden spacers, tags etc.) for any defect and carry out repair and replacement as required

PC7. maintain voltage within prescribed limits using an Off-Circuit Tap Selector (OCTS)

PC8. test for variations appearing in the primary side supply voltage and the secondary side supply voltage

PC9. check insulation resistance by using a Megger

PC10. check all loose bolts / screws / clamps, core joints, solder HT and LT terminal connections, windings for no sludge on winding to block the oil ducts and opening passage, cooling radiators, indoor and outdoor bushings for oil leakage and cracks, clasping of the conservator and any foreign items left in the tank

PC11. check all the oil gauges, dehydrating breather for saturation and color change, pressure release device and explosion vent, sealing gaskets for cracks, oil level in conservator tank gauge and thermometer, OLTC switch for arcing welding and wearing, radiator, arcing horns for dent or any defect, any rust and damage of paint on external tank, oil temperature indicator (OTI) and winding temperature indicator (WTI) for any defect and repair and replace the same if found defective

PC12. repair oil leakage and sweating in the DT and top-up oil by following instructions stated in the manual

PC13. clean the oil gauges and radiator by compressed air or water

PC14. check air-release plugs of main tank, radiator, conservator, bushings, etc. are free of air pocket / bubbles

PC15. energize DT at NO-LOAD only and check it for any abnormalities for the next 4 to 8 hours

PC16. take advice from the manufacturer or suppliers if any major abnormalities or defects found during repair and maintenance

Testing and checking of overhauled distribution transformer before delivery

To be competent, the user/individual on the job must be able to:

PC17. assemble and fit all the components of DT and packed it in its original shape

PC18. check the transformer for any leakage, low oil, silica in breather, HV & LV bushing, electrical values such as IR value (HT to E, LT to E, HT to LT, oil BDV), vent pipe is sealed with aluminum foil (diaphragm), temperature gauge is fitted and all HV terminals are fitted with horn and double screws and washers before delivery

PC19. ensure that the inspected and tested component meets the specified operating conditions before issue of OK certificate

Transformer Protection and safety

To be competent, the user/individual on the job must be able to:

PC20. Identify main components of a distribution transformer setup, including bushings, LT/HT terminals, breather, protection fuses, and lightning arresters.

PC21. Explain how over current, earth fault, and surge protection work for transformers

PC22. Detect transformer-related faults (e.g., oil leakage, bushing crack, terminal heating, fuse blown) and take initial action.

Routine Transformer Maintenance

To be competent, the user/individual on the job must be able to:

PC23. Check and maintain oil level, silica gel condition, and physical structure of transformer body.

PC24. Perform visual inspection of earthing, fencing, and lightning arresters to ensure safety.

PC25. Use hand tools and basic testing instruments (multimeter, clamp meter, megger) to assess transformer condition.

PC26. Follow personal safety protocols and use PPE (gloves, insulating shoes, helmet, etc.) while working.

Emergency Handling , Digital and Cyber security

To be competent, the user/individual on the job must be able to:

PC27. Support safe isolation during fault or maintenance using LOTO and switching procedures.

PC28. Participate in fault restoration and transformer failure response (e.g., fire, explosion, tripping).

PC29. Report any unusual behavior or fault symptoms to supervisor/control room with accurate description.

PC30. Be aware of cyber security hygiene when using mobile apps, testing tablets, or SCADA-connected DTs.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. process standards and procedures followed in the organization
KU2. how and where to keep record, accessories, gadgets, equipment, PPE's, tools & tackles systematic to maintain good house keeping
KU3. related power system aspects like ratings and various types of DT
KU4. working of a DT, its component, accessories and their functioning.
KU5. difference between dry type and oil immersed transformer and their usage at site
KU6. various material / parts / accessories required for maintenance like insulating oil, type of core, winding material, bushings, cable and conductor, cabling box, cooling radiators, conservators, oil gauges, valves, explosion vents or pressure release devices, sealing gaskets, temperature indicators, poles, insulators and fuses etc.
KU7. use of tools and kits required for testing, repair and maintenance OLTC Continuity & Resistance Measurement. (Test: Dissolve Gas Analyzer kit, temperature monitor device, Partial Discharge Test Set, Flash Point Test Set of Oil, discharge rod, chain pulley, tripod, crane, hoist, force pulley with sling, tommy bar, crimping machine, drilling machine, meggar, tong tester etc.)
KU8. procedure and technical requirements for testing, repair and maintenance of the distribution transformer
KU9. transformer winding, placing various types of insulations, fitting of core joints and complete assembly
KU10. heat treatment methods, temperature control in oven and operation of vacuum chamber
KU11. how to test the performance and condition of distribution transformer
KU12. Structure and function of power distribution transformers and their protective accessories.
KU13. Transformer protection methods like HT fuse, lightning arrester, surge diverter, and earthing system
KU14. Signs of transformer failure (oil leak, noise, discoloration, smell, tripping) and basic preventive measures.
KU15. Routine transformer maintenance practices: oil check, silica gel check, tightening terminals, cleaning.
KU16. Electrical safety rules, PPE, and site protocols for transformer yard safety.
KU17. Use of tools like IR tester, clamp meter, megger, and correct measurement recording.
KU18. Step-by-step actions during emergency or fire incident at distribution transformer site.
KU19. Role of communication: how to log, describe, and escalate transformer-related issues.
KU20. Basic awareness of connected systems like AMR/SCADA-enabled transformers.
KU21. Cyber security awareness donts while using mobile apps/tablets, avoiding unsafe digital actions, and reporting any digital tampering.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. write reports and observations related to work in English/regional language
GS2. attentively listen and comprehend the information given by the supervisor/team members

GS3. read and interpret the equipment operational documents

GS4. discuss task lists and job requirements with supervisor and co-workers

GS5. follow organization rule-based decision making process in consultation of the supervisor

GS6. analyze the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist

GS7. recognize a workplace problem and take suitable action to resolve it

GS8. Fill daily inspection sheets or transformer maintenance checklists.

GS9. Communicate clearly with control room or supervisor about faults or unusual conditions.

GS10. Follow step-by-step instructions for transformer O&M tasks.

GS11. Use PPE and testing equipment with care and attention to safety.

GS12. Act responsibly during emergency/fault handling situations.

GS13. Stay alert and avoid unsafe practices at transformer yards.

GS14. Maintain tool discipline, cleanliness, and personal conduct at site.

GS15. Handle mobile apps or tablet-based testing/data entry tools safely.

GS16. Be aware of risks in cyber-connected equipment and act cautiously.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Repairing of defective distribution transformer	**19**	**63**	**-**	**-**
PC1. check that all required tools and kits are calibrated and in good condition	1	2	-	-
PC2. prepare work area for the standard repairing procedure	1	2	-	-
PC3. identify and collect adequate spare parts to replace the faulty parts	1	1	-	-
PC4. take oil samples from tank bottom, tank top, and radiator to conduct the Break-Down Voltage (BDV) test	1	2	-	-
PC5. remove core and windings from the tank for visual inspection and ensure that they are properly covered, dry and placed safely after removal from tank	1	2	-	-
PC6. inspect core, primary winding, secondary winding, primary terminal connections, secondary terminal connections, insulation (fish paper, empire tape/cloth, wooden spacers, tags etc.) for any defect and carry out repair and replacement as required	1	3	-	-
PC7. maintain voltage within prescribed limits using an Off-Circuit Tap Selector (OCTS)	1	2	-	-
PC8. test for variations appearing in the primary side supply voltage and the secondary side supply voltage	1	3	-	-
PC9. check insulation resistance by using a Megger	1	1	-	-
PC10. check all loose bolts / screws / clamps, core joints, solder HT and LT terminal connections, windings for no sludge on winding to block the oil ducts and opening passage, cooling radiators, indoor and outdoor bushings for oil leakage and cracks, clasping of the conservator and any foreign items left in the tank	1	7	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check all the oil gauges, dehydrating breather for saturation and color change, pressure release device and explosion vent, sealing gaskets for cracks, oil level in conservator tank gauge and thermometer, OLTC switch for arcing welding and wearing, radiator, arcing horns for dent or any defect, any rust and damage of paint on external tank, oil temperature indicator (OTI) and winding temperature indicator (WTI) for any defect and repair and replace the same if found defective	2	8	-	-
PC12. repair oil leakage and sweating in the DT and top-up oil by following instructions stated in the manual	1	7	-	-
PC13. clean the oil gauges and radiator by compressed air or water	1	7	-	-
PC14. check air-release plugs of main tank, radiator, conservator, bushings, etc. are free of air pocket / bubbles	2	7	-	-
PC15. energize DT at NO-LOAD only and check it for any abnormalities for the next 4 to 8 hours	2	7	-	-
PC16. take advice from the manufacturer or suppliers if any major abnormalities or defects found during repair and maintenance	1	2	-	-
Testing and checking of overhauled distribution transformer before delivery	**7**	**11**	**-**	**-**
PC17. assemble and fit all the components of DT and packed it in its original shape	2	4	-	-
PC18. check the transformer for any leakage, low oil, silica in breather, HV & LV bushing, electrical values such as IR value (HT to E, LT to E, HT to LT, oil BDV), vent pipe is sealed with aluminum foil (diaphragm), temperature gauge is fitted and all HV terminals are fitted with horn and double screws and washers before delivery	3	5	-	-
PC19. ensure that the inspected and tested component meets the specified operating conditions before issue of OK certificate	2	2	-	-
Transformer Protection and safety	**8**	**5**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. Identify main components of a distribution transformer setup, including bushings, LT/HT terminals, breather, protection fuses, and lightning arresters.	2	2	-	-
PC21. Explain how over current, earth fault, and surge protection work for transformers	3	1	-	-
PC22. Detect transformer-related faults (e.g., oil leakage, bushing crack, terminal heating, fuse blown) and take initial action.	3	2	-	-
Routine Transformer Maintenance	**6**	**14**	**-**	**-**
PC23. Check and maintain oil level, silica gel condition, and physical structure of transformer body.	1	3	-	-
PC24. Perform visual inspection of earthing, fencing, and lightning arresters to ensure safety.	1	4	-	-
PC25. Use hand tools and basic testing instruments (multimeter, clamp meter, megger) to assess transformer condition.	1	5	-	-
PC26. Follow personal safety protocols and use PPE (gloves, insulating shoes, helmet, etc.) while working.	3	2	-	-
Emergency Handling , Digital and Cyber security	**6**	**11**	**-**	**-**
PC27. Support safe isolation during fault or maintenance using LOTO and switching procedures.	2	-	-	-
PC28. Participate in fault restoration and transformer failure response (e.g., fire, explosion, tripping).	-	5	-	-
PC29. Report any unusual behavior or fault symptoms to supervisor/control room with accurate description.	2	-	-	-
PC30. Be aware of cyber security hygiene when using mobile apps, testing tablets, or SCADA-connected DTs.	2	6	-	-
NOS Total	**46**	**104**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N2417
NOS Name	Operation and Maintenance of Distribution Transformer
Sector	Power
Sub-Sector	Distribution
Occupation	Operation and Maintenance - Distribution
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Guidelines will be same as that of QP

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1338.Apply basic health And safety practices for power related work	39	61	-	-	100	10
PSS/N1336.Working effectively with others	34	56	-	-	90	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
PSS/N2406.Testing and inspection of various faults in distribution transformer	22	78	-	-	100	35
PSS/N2417.Operation and Maintenance of Distribution Transformer	46	104	-	-	150	35
Total	**161**	**329**	**-**	**-**	**490**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.